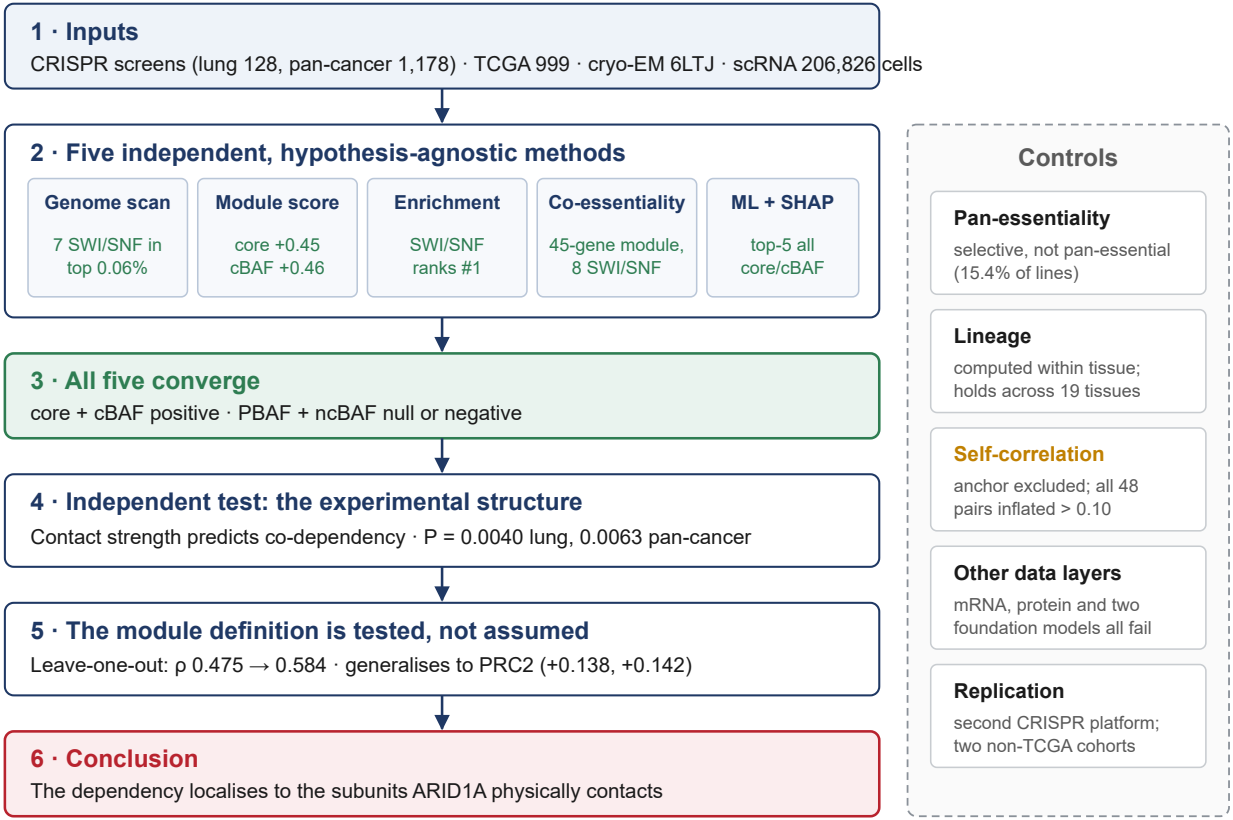

1 · Inputs
CRISPR screens (lung 128, pan-cancer 1,178) · TCGA 999 · cryo-EM 6LTJ · scRNA 206,826 cells
2 · Five independent, hypothesis-agnostic methods
Genome scan
7 SWI/SNF in top 0.06%
Module score
core +0.45 cBAF +0.46
Enrichment
SWI/SNF ranks #1
Co-essentiality
45-gene module, 8 SWI/SNF
ML + SHAP
top-5 all core/cBAF
3 · All five converge
core + cBAF positive · PBAF + ncBAF null or negative
4 · Independent test: the experimental structure
Contact strength predicts co-dependency · P = 0.0040 lung, 0.0063 pan-cancer
5 · The module definition is tested, not assumed
Leave-one-out: ρ 0.475 → 0.584 · generalises to PRC2 (+0.138, +0.142)
6 · Conclusion
The dependency localises to the subunits ARID1A physically contacts
Controls
Pan-essentiality
selective, not pan-essential (15.4% of lines)
Lineage
computed within tissue; holds across 19 tissues
Self-correlation
anchor excluded; all 48 pairs inflated > 0.10
Other data layers
mRNA, protein and two foundation models all fail
Replication
second CRISPR platform; two non-TCGA cohorts